

- In the overleaf compilation I see 25 errors
I am not looking to the latex file because I do not know about this format. In particular, it uses a special bibtex format I never used. So, I write my comments here.

Information Geometry

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- Add the context : eg DMA Genoz PhD course

YEAR(2023)

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Chapter 1

Affine Geometry

what is a mathematical space? I would say
A set where ...

1.1 Probability Simplex

Mathematical space where each point represents the probability distribution among the finite number of mutually exclusive events.

It is the set ...

Mathematically, The set of all probability measures on a finite sample space $\mathcal{X}$ is identified with the set $\mathcal{P} = \mathcal{P}(\mathcal{X})$ of all probability functions q ,

I have a memo
- set of h

$$\mathcal{P}(\mathcal{X}) = \left\{ q: \mathcal{X} \ni x \mapsto q(x) \in \mathcal{R}_+^{\mathcal{X}} \mid \sum_{x \in \mathcal{X}} q(x) = 1 \right\} \quad (1.1)$$

In the equation 1.1 every element of $q \in \mathcal{P}$ is non-negative. The set $\mathcal{P}$ is called *standard simplex* of $\mathcal{R}^{\mathcal{X}}$. It is a closed convex set of the vector space of all real functions on $\mathcal{X}$, denoted $\mathcal{R}^{\mathcal{X}}$. A *polytope*¹ is the convex hull of a finite number of points in a vector space. The generating points of the standard simplex are the delta probability functions δ_a , $a \in X$, $\delta_a(x) = (x = a)$. A *simplex* is the polytope generated by affinely independent points. All simplexes with the same number of vertexes are transformed into one another by a suitable affine function. A general

explain
affinely
independent

¹ a polytope is a geometric object with flat sides (faces) Wikipedia

explain

Chapter 1. Affine Geometry

why " - " ?

reference for convex geometry is "Barvinok [6]". The probability simplex in $\mathcal{R}^3$ can be seen in figure 1.1.

this is one of the representations of the probability simplex

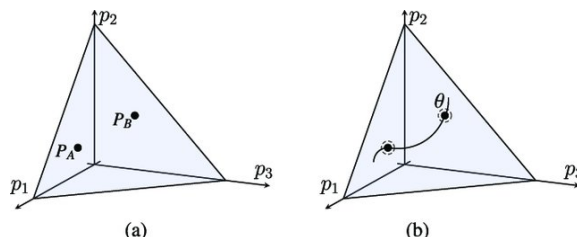


Figure 1.1:

(a) The probability simplex where the measurement has three outcomes with probabilities p_1, p_2 , and p_3 . Every probability distribution is represented by a point in the simplex (P_A and P_B). (b) θ_j parametrizes a path through the probability simplex. Sidhu and Kok [10]

Send me this reference to discuss

1.1.1 Affine Space

Some comment as before

The affine space "Barvinok [6]"².

It is a good practice to refer the section or page

of the probability simplex

$$\mathcal{A}(\mathcal{X}) = \left\{ v \in \mathcal{R}^{\mathcal{X}} \mid \sum_{x \in \mathcal{X}} v(x) = 1 \right\}. \quad (1.2)$$

It is sometimes convenient to ignore the positivity constraint for probability functions and let the "probabilities" be free to assume negative values. In the following, the key notion of *score*³.

1.1.2 Tangent Space

The *tangent space* is $TP(\mathcal{X}) = \{ v \in \mathcal{R}^{\mathcal{X}} \mid \sum_{x \in \mathcal{X}} v(x) = 0 \}$. The tangent space is the collection of all possible *velocities* of curves, that is, one-

²An n -affine simplex is the image of regular n -simplex under an affine map

³it's just the gradient of the log-likelihood function i.e. $u(\theta) = \Delta_{\theta} \log p(x|\theta)$

??
log

Refer systematically to some
text book of modern calculus of your
choice eg RUDIN or LANDAU or
other

dimensional models, in the simplex or the affine space. If the curve

$$I \ni t \mapsto q(t) \in \mathcal{A}(\mathcal{X}), \quad (1.3)$$

where I is an open real interval, is differentiable at $\bar{t}$ as a function with values in $\mathcal{R}^{\mathcal{X}}$, then

$$\dot{q}(\bar{t}) = \lim_{h \rightarrow 0} h^{-1}(q(\bar{t} + h) - q(\bar{t})) \in T\mathcal{P}(\mathcal{X}). \quad (1.4)$$

Conversely, given a time $\bar{t} \in \mathcal{R}$, a probability function $\bar{p} \in \mathcal{P}(\mathcal{X})$ and a tangent vector $v \in T\mathcal{P}(\mathcal{X})$, the curve

$$t \mapsto \bar{p} + (t - \bar{t})v \in \mathcal{A}(\mathcal{X}) \quad (1.5)$$

has derivative v at $t = \bar{t}$.

1.1.3 Acceleration of the Curve

The *acceleration* of the curve $q(\cdot)$ is defined by

$$t \mapsto \ddot{q}(t) \in T\mathcal{P}(\mathcal{X}). \quad (1.6)$$

If the acceleration is null we say that the curve is a *geodesic* ⁴

In this case, it happens if, and only if, the curve is a *mixture*, that is, $q, r \in \mathcal{A}(\mathcal{X})$ and

$$t \mapsto (1 - t)q + tr \in \mathcal{A}(\mathcal{X}). \quad (1.7)$$

Given a function $\Phi: \mathcal{A}(\mathcal{X}) \rightarrow \mathcal{R}$, the *gradient* $\text{grad } \Phi$ is a mapping from $\mathcal{A}(\mathcal{X})$ to $T\mathcal{P}(\mathcal{X})$ such that for all smooth curve $q(\cdot)$, it holds

$$\frac{d}{dt}\Phi(q(t)) = \langle \text{grad } \Phi(q(t)), \dot{q}(t) \rangle \quad (1.8)$$

^{< x, y > inner product}

⁴A locally length-minimizing curve is a geodesic. Equivalently, it is a path that a particle which is not accelerating this is would follow.

important remark. Put it
in the text and add
a reference

Chapter 1. Affine Geometry \quad \checkmark \quad \text{used in my notes}

Notice that the grad depends on the definition of velocity and on the definition of the scalar product. In this case, the componentwise computation is, given a numbering of $\mathcal{X} = [1 : n]$,

$$\begin{aligned} \frac{d}{dt} \Phi(q(1;t), \dots, q(n;t)) = \\ \sum_{j=1}^n \frac{\partial}{\partial q(j)} \Phi(q(1;t), \dots, q(n;t)) \dot{q}(j;t) = \\ \sum_{j=1}^n \left(\frac{\partial}{\partial q(j)} \Phi(\dots q(j;t) \dots) - \frac{1}{n} \sum_{k=1}^n \frac{\partial}{\partial q(k)} \Phi(\dots q(k;t) \dots) \right) \dot{q}(j;t) \end{aligned} \quad (1.9)$$

That is, $\text{grad} \neq \nabla!$

In this case, it is. In other cases don't. Explain

The previous construction provides a geometry of the probability simplex. However, *such a geometry is not the geometry used in Fisherian statistics.* You can see Part I of the monograph "Efron and Hastie [8]" for a nice presentation of the different styles of classical statistical inference, frequentist, Bayesian, and Fisherian. In turn, Fisher's methods are based on the methodology originally developed in *Genetics* and *Statistical Mechanics*. A reference to this last topic is, for example, the first part of "Landau and Lifshits [3]" Hence, we start again with a different affine geometry. We will use elementary notions of differential geometry. ~~Here is a one-slide summary~~ of the definition of a differentiable manifold. Given a set M and a real vector space V , a *chart* is a 1-to-1 mapping.

$$s: U \rightarrow V, \text{ where } U \subset M \text{ and } s(U) \text{ is open in } V. \quad (1.10)$$

Below are the two charts s_1 and s_2 in the equation 1.11

$$s_1: U_1 \rightarrow V_1 \quad \text{and} \quad s_2: U_2 \rightarrow V_2 \quad (1.11)$$

Chapter 1. Affine Geometry

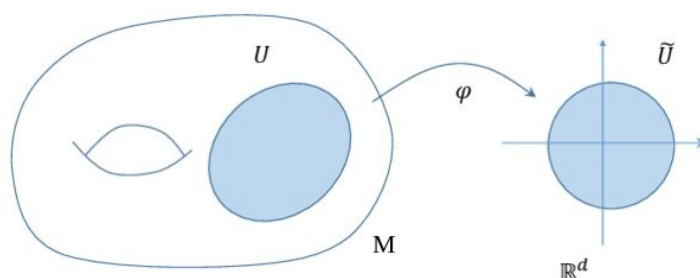
These two charts are C^k -compatible if the change-of-chart mapping

$$s_2 \circ s_1^{-1} : s_1(U_1 \cap U_2) \rightarrow s_2(U_1 \cap U_2) \quad (1.12)$$

is a C^k diffeomorphism of open sets. A family of C^k -compatible charts is a C^k -atlas. A maximal class of equivalent atlases is a *manifold*. Here we have used the ~~no~~ non-parametric version of differential geometry as presented in "Bourbaki [2] and Lang [5]"

Chose one monograph you like

1. **Manifold:** Consider a set M which has topology "Defines what are open sets and allows one to make sense of continuous maps, i.e map is continuous if the pre-image open set is open" this is not [2], [5] You are using a different reference
2. **Coordinate Chart:** Let $\phi : U \subset M \rightarrow \text{open set of } \mathbb{R}^d$, which is often denote by (U, ϕ) . Often we drop the U and just to refer ϕ for (U, ϕ) . We can see the below diagram. The definition above is more precise



→ send me this paper to discuss

Figure 1.2: Coordinate chart ϕ applied on a smooth manifold M from an open set U to $\tilde{U}$ "Zhang, Mohammadi, and Roza [13]"

In the above figure 1.3 we have only one single chart. Now we want to cover the entire manifold with coordinate patches so that the union of the domains of the definition of the coordinate charts covered the entire manifold.

The coordinate chart allows us to define some notion of differentiability in terms of the coordinate representation of s_1 which is $s_2 \circ s_1^{-1}$, but we want to be independent of the choice of chart.

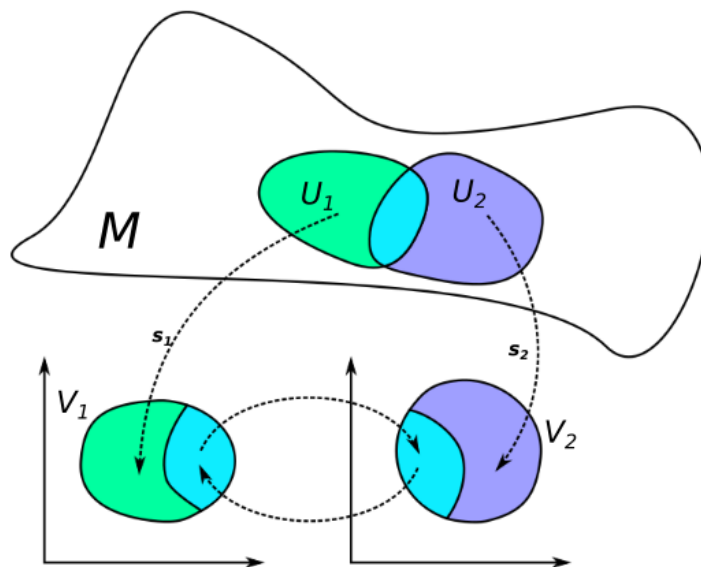


Figure 1.3: Change of Coordinate maps: Two charts are C^k -compatible if the change-of-chart mapping.

3. C^∞ **Atlas on Manifold M** is collection of charts (U_α, ϕ_α) of M such that:

- $\bigcup_\alpha U_\alpha = M$.
- For any pair of α and β with $U_\alpha \cap U_\beta \neq \emptyset$, then the $\phi_\alpha(U_\alpha \cap U_\beta)$ and $\phi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^d$ and the change of the coordinate maps $\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$ is C^∞ atlas. Now we can say that the elements of an atlas overlap smoothly.

4. Two atlases are equivalent if the $A_1 \cup A_2$ again an atlas. Given an atlas A , Let A^+ be the set of all charts (U, ϕ) such that $A \cup \{(U, \phi)\}$ is also an atlas. It is to check that A^+ is also an atlas and we called the maximal atlas or complete atlas which is generated by A . Two atlases are equivalent if they generate the same maximal atlas. So, the maximal atlas on M is a differentiable structure on M .

1.1.4 Affine Spaces

Given a set M and a real finite dimensional vector space V , an *affine space* (Weyl [1]) $(M, V, \vec{\cdot})$ consists of a *displacement* mapping

$$M \times M \ni (p, q) \mapsto \vec{pq} \in V,$$

such that

- for each p the mapping $s_p: M \ni q \mapsto s_p(q) = \vec{pq}$ is 1-to-1, and
- the *parallelogram law*, $\vec{pq} + \vec{qr} = \vec{pr}$, that is, $s_p(q) + s_q(r) = s_p(r)$, holds.

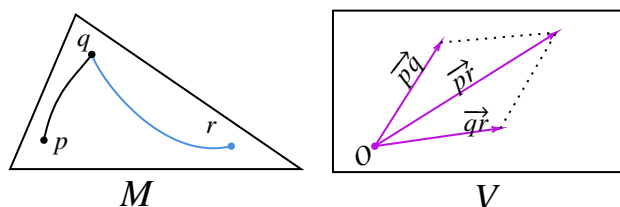


Figure 1.4: Affine space

1.1.5 Affine manifold

In particular, $\vec{pp} = s_p(p) = 0$ and $\vec{pq} + \vec{qp} = s_p(q) + s_q(p) = 0$. s_p is a *chart* if the image is open. The *atlas of charts* $s_p: M \rightarrow V, p \in M$, defines

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an *affine manifold*. An affine manifold is a manifold, but the atlas defining the affine manifold is made up of special functions "Chirco and Pistone [12]" All change-of-charts mappings of the atlas are vector translations associated with a change of origin:

$$s_p \circ s_q^{-1} : v \mapsto s_p(q) + v \quad (1.13)$$

If $t \mapsto \rho(t) \in M$ is a curve, for each origin p the curve $t \mapsto s_p(\rho(t))$ is V -valued. The derivative $\frac{d}{dt}s_p(\rho(t))$ does not depend on p because

$$\frac{d}{dt}s_p(\rho(t)) = \frac{d}{dt}(s_p(q) + s_q(\rho(t))) = \frac{d}{dt}s_q(\rho(t)) \quad (1.14)$$

$t\rho(t) = \frac{d}{dt}s_p(\rho(t))$ is, by definition, the *affine velocity*.

Example 1: Let us consider the base set M which is probability simplex $\mathcal{P}(\mathcal{X})$ and the vector space V is the tangent space of the simplex,

$$V = T\mathcal{P}(\mathcal{X}) = \left\{ v \in \mathbb{R}^{\mathcal{X}} \left| \sum_{x \in \mathcal{X}} v(x) = 0 \right. \right\}. \quad (1.15)$$

The displacement is $\overrightarrow{pq} = q - p \in V$ because $\sum_x (q(x) - p(x)) = 1 - 1 = 0$. The mapping $s_p(q) = q - p$ is 1-to-1. The parallelogram law is $(q - p) + (r - q) = (r - p)$. A *vector base* of V produces a *faithful parametrization* of M . For example, a base of $T\{1, 2, 3\}$ is $v_1 = (1, -1, 0)$, $v_2 = (1, 0, -1)$, and

$$(p, q) \mapsto s_p(q) = -(q(2) - p(2)) \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - (q(3) - p(3)) \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \mapsto$$

$$(-(q(2) - p(2)), -(q(3) - p(3))) \in$$

$$\text{hull}((1, 0), (1, -1), (0, -1), (-1, 0), (-1, 1), (0, 1)) \subset \mathbb{R}^2 \quad (1.16)$$

explains

check this

Example 2: Open Probability Simplex In the previous example, the image of each $s_p: q \mapsto q - p$ is a translation by $-p$ of the simplex. Hence it is closed in the tangent space, and the manifold structure is not well defined. We need an open image. Let us take as the base set the open simplex

$$\mathcal{P}_>(\mathcal{X}) = \{q \in \mathcal{P}(\mathcal{X}) \mid q(x) > 0, x \in \mathcal{X}\} \quad (1.17)$$

and the same vector space as in the previous case,

$$V = T\mathcal{P}(\mathcal{X}) = T\mathcal{P}_>(\mathcal{X}) \quad (1.18)$$

The displacement $s_p(q) = q - p$ satisfies the parallelogram law, and the image of s_p is $\mathcal{P}_>(\mathcal{X}) - p$ hence it is open in the tangent space. Hence, s_p is a chart. The inverse is $s_p^{-1}(v) = v + p$ and the change-of-chart is

$$v \mapsto s_p^{-1}(v) = v + p \mapsto s_q \circ s_p^{-1}(v) = (v + p) - q = v + s_q(p) \quad (1.19)$$

In the practice of Statistical Physics and Fisher's Statistics, other affine geometries imposed themselves Lang [5] *why cite Lang here?*

1.1.6 Fisher Information

Let $X_1, X_2, \dots, X_n$ are random samples. Mathematically,

$$I(\theta) = E \left\{ \frac{\partial^2}{\partial \theta^2} \ell(X|\theta) \right\} \quad (1.20)$$

expand or drop

1.1.6.1 Fisher Score

Fisher Score in information geometry is defined by Aitchison's geometry by using a richer atlas of charts. There is a relation between Fisher's arguments and Statistical Physics.

explain or drop

Chapter 1. Affine Geometry

A *curve* or *one-dimensional model* is a mapping

$$I \ni \theta \mapsto q(\theta) \in X \quad (1.21)$$

As $\mathcal{P}(\mathcal{X}) \subset \mathbb{R}^{\mathcal{X}}$, we can assume I open and the mapping differentiable. In such a case,

$$\dot{q}(t) = \lim_{h \rightarrow 0} h^{-1}(q(t+h) - q(t)) \in T\mathcal{P}(\mathcal{X}) . \quad (1.22)$$

Precisely because of that, we use the term “tangent space”: the definition of tangent space depends on the definition of velocity of variation of the curve.

Let us suppose that there exists a couple $(x, \theta_0) \in \mathcal{X} \times I$ such that the curve $\theta \mapsto q(\theta)$ hits the facet $\{p \mid p(x) = 0\}$ at $\theta = \theta_0$, that is $q(x; \theta_0) = 0$. The differentiable curve $\theta \mapsto q(x, \theta)$ has a minimum at θ_0 , hence $\dot{q}(x, \theta_0) = 0$.

It holds $q(x; \theta) = 0 \Rightarrow \dot{q}(x; \theta) = 0$, hence and exists a curve $t \mapsto \dot{q}^*(t) \in \mathbb{R}^{\mathcal{X}}$, the *Fisher's score*, such that

$$\dot{q}(x; \theta) = \dot{q}^*(x; \theta)q(x; \theta) . \quad (1.23)$$

It is an *absolute continuity* setting, $\dot{q}(\theta) \ll q(\theta)$ "Ay et al. [9]". In particular, if $\theta \mapsto q(\theta) \in \mathcal{P}_{>}\mathcal{X}$, then the *Fisher's score* is $\dot{q}^*(\theta) = \frac{d}{d\theta} \log q(\theta)$. There are many statistical reasons to take the Fisher score as a useful notion of the velocity of variation in the parameters of a statistical model ("Efron and Hastie [8] and Ibragimov and Hasminskii [4]"). *explain*

1.2 Statistical Bundle, Contrast Bundle

Let us consider for each $q \in \mathcal{P}_{>\mathcal{X}}$ the vector space of q -contrasts, that is, the random variables which are centered for q ,

$$B_q = \left\{ u \in \mathbb{R}^{\mathcal{X}} \left| \sum_{x \in \mathcal{X}} u(x)q(x) = 0 \right. \right\}. \quad (1.24)$$

The *statistical bundle* of $\mathcal{P}_{>\mathcal{X}}$ is the bundle $S\mathcal{P}_{>\mathcal{X}} \rightarrow \mathcal{P}_{>\mathcal{X}}$ of spaces of contrasts,

$$S\mathcal{P}_{>\mathcal{X}} = \left\{ (q, v) \in \mathcal{P}_{>\mathcal{X}} \times \mathbb{R}^{\mathcal{X}} \left| \mathbb{E}_q[u] = 0 \right. \right\} \quad (1.25)$$

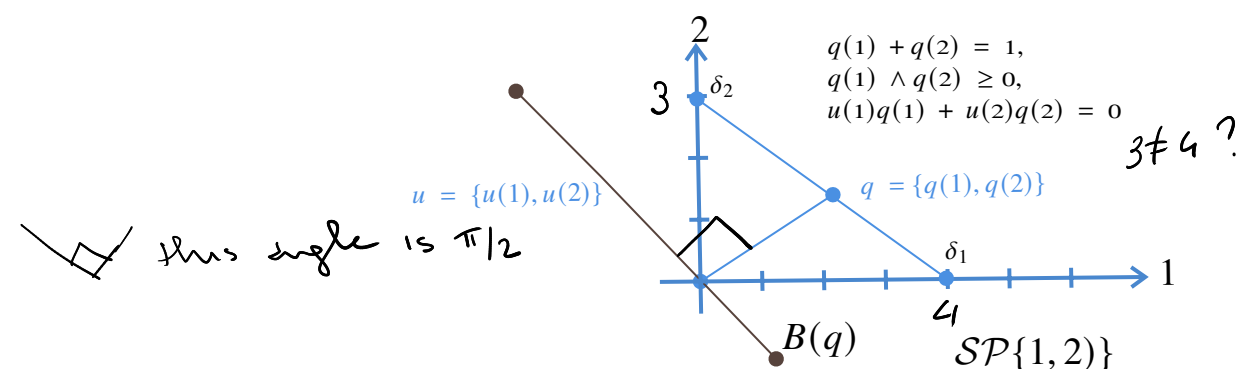


Figure 1.5: Statistical Bundle

1.3 Parallel Transport

In differential geometry, affine connection ∇ defines a way to “translate” vectors = “parallel transport” vectors on tangent planes along a smooth curve $c(t)$. (Manifolds are not embedded) Parallel transport of vectors between tangent planes T_p and T_q linked by c (defined by ∇), with

explain

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$c(0) = p$ and $c(1) = q$:

$$\forall v \in T_p = \prod_{c(0) \mapsto c(1)}^{\nabla} v \in T_{c(t)} \quad (1.26)$$

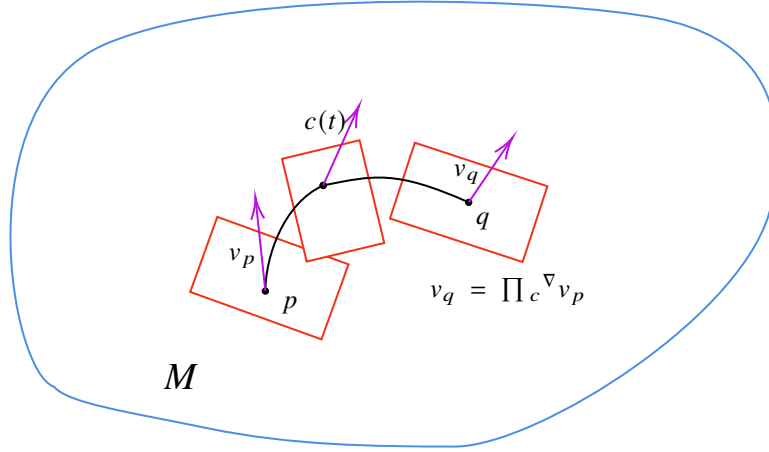


Figure 1.6: Parallel Transport

Each fiber B_q represents the tangent vectors (velocities) at q . We need to introduce "connections" between different "expressions" of the tangent space. Each fiber B_q has an *inner product* $\langle u, v \rangle_q = \mathbb{E}_q [uv]$. The mapping

$$S^2\mathcal{P}_{>\mathcal{X}} \ni (q, u, v) \mapsto \langle u, v \rangle_q \quad (1.27)$$

is called the *metric* of the statistical bundle.

Definition: For all $(p, u), (q, v) \in S\mathcal{P}_{>\mathcal{X}}$,

- ${}^e\mathbb{U}_p^q: B_p \ni u \mapsto u - \mathbb{E}_q [u] \in B_q$ is the *exponential transport*;
- ${}^m\mathbb{U}_q^p: B_q \ni v \mapsto \frac{q}{p} v \in B_p$ is the *mixture transport*.

Duality: $\forall (p, u), (q, v) \in S\mathcal{P}_{>\mathcal{X}}$,

$$\langle u, {}^m\mathbb{U}_q^p v \rangle_p = \langle {}^e\mathbb{U}_p^q u, v \rangle_q \quad (1.28)$$

Proof:

$$\begin{aligned} \langle u, {}^m\mathbb{U}_q^p v \rangle_p &= \left\langle p, \frac{q}{p} v \right\rangle_p = \mathbb{E}_p \left[uv \frac{q}{p} \right] = \\ \mathbb{E}_q [uv] &= \mathbb{E}_q [(u - \mathbb{E}_q[u])v] = \langle u - \mathbb{E}_q[u], v \rangle_q = \langle {}^e\mathbb{U}_p^q u, v \rangle_q \end{aligned} \quad (1.29)$$

1.3.1 Affine Spaces with Parallel Transport

Because of what was said about the notion of Fisher's score and the bundle of contrasts, it is useful to rephrase Weyl's definition by moving from the trivial tangent bundle $M \times V$ to the nontrivial bundle of contrasts or fluctuations.⁵

Definition: Affine Space Let M be a set and $B_p, p \in M$, a family of topological vector spaces (top-linear spaces). Let $({}^0\mathbb{U}_p^q), p, q \in M$ be a cocycle of top-linear isomorphism ${}^0\mathbb{U}_p^q: B_p \rightarrow B_q$, ${}^0\mathbb{U}_q^p {}^0\mathbb{U}_p^q = I_{B_p}$. Define a displacement mapping

$$\mathbb{S}: M \times M \ni (p, q) \mapsto s_p(q) \in B_p \quad (1.30)$$

so that:

1. For each fixed p the mapping $s_p: q \mapsto s_p(q) = \mathbb{S}(s, p)$ is *injective*
2. (Parallelogram law) $\mathbb{S}(p, q) + {}^0\mathbb{U}_q^p \mathbb{S}(q, r) = \mathbb{S}(p, r)$

⁵While still staying in the finite state space case, I start using a more general language taken from infinite-dimensional differential geometry. See Lang [5]

1.4 Affine manifold

The affine space provides an atlas of charts $s_p: M \rightarrow B_p$, $p \in M$. The *change-of-chart* map is $s_p \circ s_q^{-1}$. At $\rho = s_q^{-1}(w)$, $w \in B_q$, it holds

$$s_p \circ s_q^{-1}(w) = s_p(\rho) = s_p(q) + {}^0\mathbb{U}_q^p s_q(\rho) = s_p(q) + {}^0\mathbb{U}_q^p w. \quad (1.31)$$

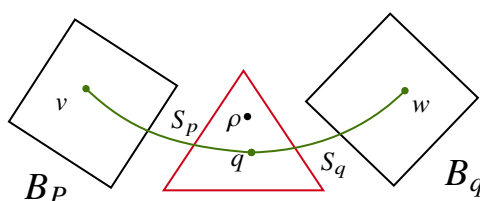


Figure 1.7: Affine Manifold

The change-of-origin map is the restriction of an affine map whose linear part is the parallel transport.

Assume that the vector fibers of the affine space $(M, (B_p p), \mathbb{S})$ are Banach spaces and assume that for each p , $s_p M$ is a neighborhood of 0 in B_μ . Define $U_p = s_p^{-1}(s_p(M)^\circ)$. Then $(s_p: U_p)$ is a chart on M . The charts are compatible, and the resulting manifold is the *affine manifold* of the affine space.

1.5 Affine Bundle

The specific form of the atlas defining the affine manifold allows the extension of the same atlas to define an affine bundle.

Given the affine manifold, consider the set

$$\{(p, v) \mid p \in M, v \in B_q\} \quad (1.32)$$

and, for each $p \in M$ define the chart

$$s_p(q, v) = (s_p(q), {}^0\mathbb{U}_p^q v) \in B_p \times B_p \quad (1.33)$$

to define the manifold S .

The statistical bundle of the affine manifold is where we define the velocity of a curve. The affine bundle has a tangent bundle where we define the second-order geometry that is, accelerations.

1.6 Exponential affine space

- We define the exponential displacement on $\mathcal{P}_>(\mathcal{X})$ by

$$\mathcal{P}_>(\mathcal{X}) \times \mathcal{P}_>(\mathcal{X}) \ni (p, q) \mapsto s_p(q) = \log \frac{q}{p} - \mathbb{E}_p \left[\log \frac{q}{p} \right] \in B_p, \quad (1.34)$$

and the exponential transport by

$${}^e\mathbb{U}_p^q: B_p \ni v \mapsto v - \mathbb{E}_q[v] \in B_q. \quad (1.35)$$

- The (generalized) parallelogram law is

$$\begin{aligned} \left(\log \frac{q}{p} - \mathbb{E}_p \left[\log \frac{q}{p} \right] \right) + {}^e\mathbb{U}_q^p \left(\log \frac{r}{q} - \mathbb{E}_q \left[\log \frac{r}{q} \right] \right) = \\ \left(\log \frac{q}{p} - \mathbb{E}_p \left[\log \frac{q}{p} \right] \right) + \left(\log \frac{r}{q} - \mathbb{E}_p \left[\log \frac{r}{q} \right] \right) = \\ \log \frac{r}{p} - \mathbb{E}_p \left[\log \frac{r}{p} \right]. \end{aligned} \quad (1.36)$$

- The inverse chart is defined on all of B_p by

$$\begin{aligned} s_p^{-1}(v) = \exp\{v - K_p(v)\} \cdot p = e_p(v) \\ K_p(v) = \log \mathbb{E}_p[e^v]. \end{aligned} \quad (1.37)$$

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- The *cumulant functional* K_p has several important well-known properties.

$$1. \ D(p \parallel q) = \mathbb{E}_p \left[\log \frac{p}{q} \right] = \mathbb{E}_p \left[\log \frac{p}{\exp\{v - K_p(v)\} \cdot p} \right] = K_p(v),$$

that is,

$$K_p(s_p(q)) = D(p \parallel q) . \quad (1.38)$$

$$2. \ dK_p(v)[h] = \mathbb{E}_{e_p(v)}[h] = \left\langle \frac{e_p(v)}{p} - 1, h \right\rangle_{e_p(v)}$$

$$3. \ d^2 K_p(v)[h, k] = \text{Cov}_{e_p(v)}(h, k) = \left\langle e\mathbb{U}_p^{e_p(v)} h, e\mathbb{U}_p^{e_p(v)} k \right\rangle_{e_p(v)} = \left\langle h, \left(m\mathbb{U}_{e_p(v)}^p e\mathbb{U}_p^{e_p(v)} \right) k \right\rangle_p .$$

+ Add proofs

1.7 Mixture affine space

- We define the *mixture displacement* on $\mathcal{P}_{>}(\mathcal{X})$ by

$$\mathcal{P}_{>}(\mathcal{X}) \times \mathcal{P}_{>}(\mathcal{X}) \ni (p, q) \mapsto \eta_p(q) = \frac{q}{p} - 1 \in B_p ,$$

and the *mixture transport* by

$$m\mathbb{U}_p^q : B_p \ni v \mapsto \frac{p}{q} v \in B_q . \quad (1.39)$$

- The (generalized) parallelogram law is

$$\left(\frac{q}{p} - 1 \right) + \frac{q}{p} \left(\frac{r}{q} - 1 \right) = \left(\frac{r}{p} - 1 \right) .$$

- The inverse chart is defined for all $v > 1$, $w \in B_p$ by

$$\eta_p^{-1}(v) = (1 + v) \cdot p . \quad (1.40)$$

1.8 Aitchison geometry

If the base set is the positive probability simplex $\mathcal{P}_>(\mathcal{X})$ with tangent space $T\mathcal{P}_>(\mathcal{X})$, define the displacement mapping $s_p: \mathcal{P}_>(\mathcal{X}) \rightarrow T\mathcal{P}_>(\mathcal{X})$ by "Erb and Ay [11]"

$$s_p(q) = \log \frac{q}{p} - \sum_{y \in X} \frac{1}{\#X} \log \frac{q(y)}{p(y)} = \log \frac{q}{p} - \mathbb{E} \left(\log \frac{q}{p} \right) . \quad (1.41)$$

The difference with the exponential charts is the centering that does not depend on p . The mapping s_p is surjective, with inverse

$$s_p^{-1}(u) = \exp () (u - K(u)) \cdot p$$

$$K(u) = \log \sum_{y \in X} e^{u(y)} p(y) = \log \mathbb{E}_p [e^u] . \quad (1.42)$$

Notice the relationship between the two forms of the normalizing constant:

$$K(u) = \mathbb{E} \left(\log \frac{p}{q} \right) . \quad (1.43)$$

The parallelogram law holds:

$$\left(\log \frac{q}{p} - \mathbb{E} \left(\log \frac{q}{p} \right) \right) + \left(\log \frac{r}{q} - \mathbb{E} \left(\log \frac{r}{q} \right) \right) = \log \frac{r}{p} - \mathbb{E} \left(\log \frac{r}{p} \right) \quad (1.44)$$

If p is uniform, that is, $p(x) = D^{-1}$, $D = \#X$, then

$$s_{D^{-1}}(q)(x) = \log q(x) - \sum_{y \in X} \log q(y) = \log \frac{q(x)}{(\prod_{y \in X} q(y))^{1/\#X}} \quad (1.45)$$

Notice that this special chart is equal to a chart in the exponential atlas. The exponential affine and Aitchison geometry are equal, just presented

in different atlases. Precisely, the expression of the tangent space is the statistical bundle in one case and the tangent space of the simplex in the other case.

In Aitchison's geometry, the coordinates given by the chart above are called *centered log-ratio coefficients* clr "Pawlowsky-Glahn, Egozcue, and Tolosana-Delgado [71]".

Any n -sequence (n -sample) $q_1, \dots, q_n \in \mathcal{P}_>(X)$ is mapped to an n -sequence in the space of coordinates (contrasts),

$$u_1 = \text{clr}(q_1), \dots, u_n = \text{clr}(q_n) \in T\mathcal{P}_>(X). \quad (1.46)$$

The space of contrasts $T\mathcal{P}_>(X)$ is a vector space with an inner product $\langle u, v \rangle = \mathbb{E}(uv) = \text{Cov}(u, v)$ so that we can perform all those computations that require a vector space structure.

The sample mean is

$$\begin{aligned} \frac{1}{n} \sum_{j=1}^n u_j &= \frac{1}{n} \sum_{j=1}^n \text{clr}(q_j) = \\ &= \frac{1}{n} \sum_{j=1}^n \left(\log q_j - \sum_{y \in X} \log q_j(y) \right) = \\ &= \log \left(\prod_{j=1}^n q_j \right)^{1/n} - \sum_{y \in X} \log \left(\prod_{j=1}^n q_j(y) \right)^{1/n} = \\ &= \text{clr} \left(\frac{\left(\prod_{j=1}^n q_j \right)^{1/n}}{\mathbb{E} \left(\left(\prod_{j=1}^n q_j \right)^{1/n} \right)} \right) \quad (1.47) \end{aligned}$$

The sample variance is based on the squared norm

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$$\begin{aligned} \langle \text{clr}(q), \text{clr}(p) \rangle &= \mathbb{E}(\text{clr}(q) \text{clr}(r)) = \\ &= \mathbb{E}(\log q \log r) - \mathbb{E}(\log q) \mathbb{E}(\log r) = \text{Cov}(\log q, \log r) . \end{aligned} \quad (1.48)$$

The velocity is

$$\frac{D}{dt}q(t) = \frac{d}{dt} \text{clr}(q(t)) = \frac{\dot{q}(t)}{q(t)} - \mathbb{E}\left(\frac{\dot{q}(t)}{q(t)}\right) \quad (1.49)$$

The velocity curve $t \mapsto \frac{D}{dt}q(t)$ is a curve in a vector space. Its derivative is the *acceleration*

$$\frac{d}{dt} \frac{D}{dt}q(t) = \frac{\ddot{q}(t)}{q(t)} - \left(\frac{\dot{q}(t)}{q(t)}\right)^2 - \mathbb{E}\left(\frac{\ddot{q}(t)}{q(t)} - \left(\frac{\dot{q}(t)}{q(t)}\right)^2\right) \quad (1.50)$$

Notice that the velocity is a deviation from a mean value in the uniform probability. The corresponding intensity is the variance of $\frac{d}{dt} \log q(t)$ in the uniform probability. *This choice does not fit Fisher's method, see below.*

The curves of zero acceleration are the curves of constant velocity. We call such curves *geodesics*. A geodesic from $p = \text{clr}(u)$ and initial velocity v has the form

$$q(t) = \text{clr}^{-1}(tv + u) = \exp\{tu - K_p(u)\} \cdot p . \quad (1.51)$$

Given a definition of the inner product in the vector space of an affine space, the *gradient* of a real function Φ on the base set (scalar field) is a mapping $\text{grad } \Phi$ from the base set to the vector space, the tangent space, that is, a vector field, such the for all smooth curve q it holds

$$\frac{d}{dt}\Phi(q(t)) = \left\langle \text{grad } \Phi(q(t)), \frac{D}{dt}q(t) \right\rangle . \quad (1.52)$$

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In Aitchison geometry, if $\text{clr}(q(t)) = u(t)$ and $q(t) = \text{clr}^{-1}(u(t))$, then

$$\begin{aligned} \frac{d}{dt} \Phi \circ \text{clr}^{-1}(u(t)) &= \mathbb{E} \left(\text{grad } \Phi \circ \text{clr}^{-1}(u(t)) \dot{u}(t) \right) = \\ &= \frac{1}{\#X} \sum_{x \in X} \text{grad } \Phi \circ \text{clr}^{-1}(u(x; t)) \dot{u}(x; t) \end{aligned} \quad (1.53)$$

It follows that

$$\frac{\partial}{\partial u(x)} \Phi \circ \text{clr}^{-1}(u) = \frac{1}{\#X} \text{grad } \Phi \circ \text{clr}^{-1}(u)(x) \quad (1.54)$$

or

$$\frac{1}{\#X} \text{grad } \Phi(q) = \nabla \left(\Phi \circ \text{clr}^{-1} \right) \circ \text{clr} \quad (1.55)$$

1.9 Exercises

Example 1: Gradient of the Entropy The entropy is defined for $q \in \mathcal{P}_{>}(\mathcal{X})$ by

$$\mathcal{H}(q) = \mathbb{E}_q[-\log q] = \sum_{x \in X} -\log q(x) q(x)$$

As $-\log$ is strictly convex, and $q(x) < 1$,

$$0 < -\log(\mathbb{E}_q[q]) < \mathcal{H}(q) .$$

As $s(x) = -x \log x$ is strictly concave,

$$\frac{1}{\#X} \mathcal{H}(q) = \sum_{x \in X} \frac{1}{\#X} s(q(x)) \leq s\left(\sum_{x \in X} \frac{1}{\#X} q(x)\right) = \frac{1}{\#X} \log \#X .$$

If m is the uniform probability function,

$$\mathcal{H}(m) = \sum_{x \in X} \frac{1}{\#X} \log\left(\frac{1}{\#X}\right) = \log \#X .$$

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The entropy function is maximal at the uniform probability function.

By definition of *gradient* of a scalar field $\Phi: \mathcal{P}_>(\mathcal{X}) \rightarrow \mathbb{R}$, for each smooth curve $t \mapsto q(t)$,

$$\begin{aligned}
 \left\langle \text{grad } \mathcal{H}(q(t)), \frac{D}{dt} q(t) \right\rangle_{q(t)} &= \frac{d}{dt} \mathcal{H}(q(t)) \\
 &= \frac{d}{dt} \sum_{x \in X} s(q(x; t)) \\
 &= \sum_{x \in X} s'(q(x; t)) \frac{d}{dt} q(x; t) \\
 &= \sum_{x \in X} (-1 - \log(q(x; t))) \frac{D}{dt} q(x; t) q(x; t) \\
 &= \mathbb{E}_{q(t)} \left[-(1 + \log(q(t))) \frac{D}{dt} q(t) \right]
 \end{aligned}$$

The gradient must be centered so that

$$\text{grad } \mathcal{H}(q) = -(1 + \log(q)) - \mathbb{E}_q[-(1 + \log(q))] = \boxed{-\log(q) - \mathcal{H}(q)}$$

Notice that $\text{grad } \mathcal{H}(m) = 0$.

The *gradient flow equation* is

$$\boxed{\frac{D}{dt} q(t) = -\text{grad } \mathcal{H}(q(t))}$$

which is a *replicator equation* "Ay et al. [9]"

$$\frac{d}{dt} q(x; t) = q(x; t) \left(-\log(q(x; t)) - \sum_{y \in X} -\log(q(y; t)) q(y; t) \right).$$

If $q(0) = q_0$, then $\dot{q}^*(0) = \frac{D}{dt} q(t)|_{t=0} = -\log(q_0) - \mathcal{H}(q_0)$.

A computation shows that the solution of the gradient flow is a (non-canonical) exponential family of the form

$$q(t) = \exp\{\theta(t)\dot{q}^*(0) - K_{q_0}(\theta(t)\dot{q}^*(0))\} \cdot q_0.$$

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with $\theta(0) = 0$

The first step is to compute the velocity. From

$$\log q(t) = \theta(t)\dot{q}^*(0) - K_{q_0}(\theta(t)\dot{q}^*(0)) + \log q_0$$

and the equation for the derivative of the cumulant function, the velocity is

$$\dot{q}^*(t) = \dot{\theta}(t)\dot{q}^*(0) - \mathbb{E}_{q(t)} [\dot{\theta}(t)\dot{q}^*(0)] = \dot{\theta}(t) (-\log q_0 + \mathbb{E}_{q(t)} [\log q_0])$$

The entropy is

$$\begin{aligned} \mathcal{H}(q(t)) &= \mathbb{E}_{q(t)} [-\log q(t)] = \\ &\quad \mathbb{E}_{q(t)} [-\theta(t)\dot{q}^*(0) - \log q_0] + K_{q_0}(\theta(t)\dot{q}^*(0)) \end{aligned}$$

The gradient of the entropy is

$$\begin{aligned} -\log q(t) - \mathcal{H}(q(t)) &= \\ &\quad \theta(t)(-\dot{q}^*(0) + \mathbb{E}_{q(t)} [\dot{q}^*(0)]) - \log q_0 + \mathbb{E}_{q(t)} [\log q_0] = \\ &\quad (\theta(t) - 1) (\log q_0 - \mathbb{E}_{q(t)} [\log q_0]) \end{aligned}$$

The gradient flow equation is satisfied by

$$\dot{\theta}(t) = \theta(t) - 1 \Rightarrow \theta(t) = e^t - 1$$

which in turn gives

$$q(t) \propto q_0^{e^t}.$$

It is also interesting to compute the limits $t \rightarrow \pm\infty$. Precisely, $\lim_{t \rightarrow -\infty} q(t) = m$, and the other case follows from

$$\left(\frac{q_0(x)}{\max_x q_0(x)} \right)^{e^t} \rightarrow \begin{cases} 1 & \text{if } q_0(x) = \max_x q_0(x) \\ 0 & \text{if } q_0(x) < \max_x q_0(x) \end{cases}$$

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Fisher Information

$\theta \mapsto q(\theta)$ smooth curve $\dot{q}^*(\theta) = \frac{d}{dt} \log q(\theta)$ and $\mathbb{E}_{q(\theta)} [\dot{q}^*(\theta)] = 0$

Fisher Information:-

$$\mathcal{I}(\theta) = \text{var}\{\dot{q}^*(\theta)\} = \mathbb{E}_{q(\theta)} [\dot{q}^*(\theta)^2]$$

Gradient of Φ :- $\Phi : \mathcal{P}_>(\mathcal{X}) \mapsto \mathbb{R}$

$$\frac{d}{dt} \Phi(q(\theta)) = \langle \text{grad } \Phi(q(\theta)), \dot{q}^*(\theta) \rangle_{q(\theta)}$$

Cramer-Rao inequality:-

$$\begin{aligned} \left\{ \frac{d}{dt} \Phi(q(\theta)) \right\}^2 &= \{ \langle \text{grad } \Phi(q(\theta)), \dot{q}^*(\theta) \rangle_{q(\theta)} \}^2 \\ &\leq \mathbb{E}_{q(\theta)} [\text{grad } \Phi(q(\theta))]^2 \mathbb{E}_{q(\theta)} [\dot{q}^*(\theta)^2] \end{aligned}$$

$$\boxed{\mathbb{E}_{q(\theta)} [\text{grad } \Phi(q(\theta))]^2 \geq \left\{ \frac{d}{d(\theta)} \Phi(q(\theta)) \right\}^2 \mathcal{I}(\theta)^{-1}}$$

Example 2: Fisher Information $\Phi(q) = \mathbb{E}_q [t]$ and " t " is an unbiased estimate of $\Phi(q)$.

The error is given by

$$\text{grad } \Phi(q) = t - \mathbb{E}_q [t] = t - \Phi(q)$$

The error variance will be

$$\mathbb{E}_q [\{\text{grad } \Phi(q)\}^2] = \text{var}_q(t)$$

The Cramer-Rao inequality is

$$\text{var}_{q(\theta)}(t) \geq \left\{ \frac{d}{d(\theta)} \Phi(q(\theta)) \right\}^2 \mathcal{I}(\theta)^{-1}$$

} check on literature
eg
Efron Heston

Example 2.1 The entropy is

$$\Phi(q) = -\mathbb{E}_q [\log(q)]$$

In the subject of statistical physics

$$\text{grad } \Phi(q) = -\log(q) - \{-\mathbb{E}_q [\log(q)]\}$$

The Cramer-Rao inequality is

$$\text{var}_{q(\theta)} \log(q(\theta)) \geq \left\{ \frac{d}{d(\theta)} \Phi(q(\theta)) \right\}^2 \mathcal{I}(\theta)^{-1}$$

Exercise: Replicator equation

Cf the literature with that keyword "Ay et al. [9]"

Let be given a real function $f \in C^1$ and consider the function of $\mathcal{P}_{>\mathcal{X}}$ defined by

$$H_f(q) = \mathbb{E}_q [f \circ q] = \sum_{x \in \mathcal{X}} q(x) f(q(x))$$

For example, the entropy is $\mathcal{H}(q) = H_{-\log}(q)$. There are other entropies of interest. The Kullback-Leibler divergence (from $p \in \mathcal{P}_{>\mathcal{X}}$) is

$$q \mapsto D(q \| p) = H_f(q), \quad f(q) = \log \frac{q}{p}.$$

The Tsallis entropy is defined for $\alpha \in \mathbb{R}, \alpha \neq 1$ by

$$S_\alpha(q) = \frac{1}{\alpha - 1} \left(1 - \sum_{x \in X} q(x)^\alpha \right) = \mathbb{E}_q \left[\frac{1 - q^{\alpha-1}}{\alpha - 1} \right] = H_f(q)$$

with $f(q) = \frac{1-q^{\alpha-1}}{\alpha-1}$. Note $\lim_{\alpha \rightarrow 1} \frac{1-q^{\alpha-1}}{\alpha-1} = -\log q$.

Other entropies, such as Renyi entropy are of a more general form.

Let us compute the gradient of H_f . By definition of gradient of a scalar field $\Phi: \mathcal{X} \rightarrow \mathbb{R}$, for each smooth curve $t \mapsto q(t)$, the inner product

$\langle \text{grad } H_f(q(t)), \frac{D}{dt}q(t) \rangle_{q(t)}$ equals

$$\begin{aligned} \frac{d}{dt}H_f(q(t)) &= \frac{d}{dt} \sum_{x \in \mathcal{X}} q(x;t) f(q(x;t)) \\ &= \sum_{x \in \mathcal{X}} \dot{q}(x;t) f(q(x;t)) + q(x;t) f'(q(x;t)) \dot{q}(x;t) \\ &= \sum_{x \in \mathcal{X}} (f(q(x;t)) + q(x;t) f'(q(x;t))) \frac{D}{dt}q(x;t) q(x;t) \\ &= \mathbb{E}_{q(t)} \left[(f(q(x;t)) + q(x;t) f'(q(x;t))) \frac{D}{dt}q(t) \right] \end{aligned}$$

Let us define $F(q) = f(q) + q f'(q)$. The gradient is

$$\text{grad } H_f(q) = F(q) - \mathbb{E}_q [F(q)]$$

and the gradient flow equation of H_f is the replicator equation

$$\frac{D}{dt}q(t) = F(q(t)) - \mathbb{E}_{q(t)} [F(q(t))] .$$

If the initial position is $q(0) = p$, then $\dot{q}^*(0) = F(p) - H_F(p)$. ⁶

1.10 Exercises: Deformed Logarithm

The mechanism of affine spaces can be generalized to deformed logarithms.

Let us consider that mapping $A : R_t \mapsto R_t$ is non-decreasing and let us

define the $\log_A x = \int_1^x \frac{du}{A(u)}$ and is a concave function.

Example 1:-

Let $A(u) = u$ and $\log x = \int_1^x \frac{du}{u}$ This is the ordinary log. **Example 2:-**

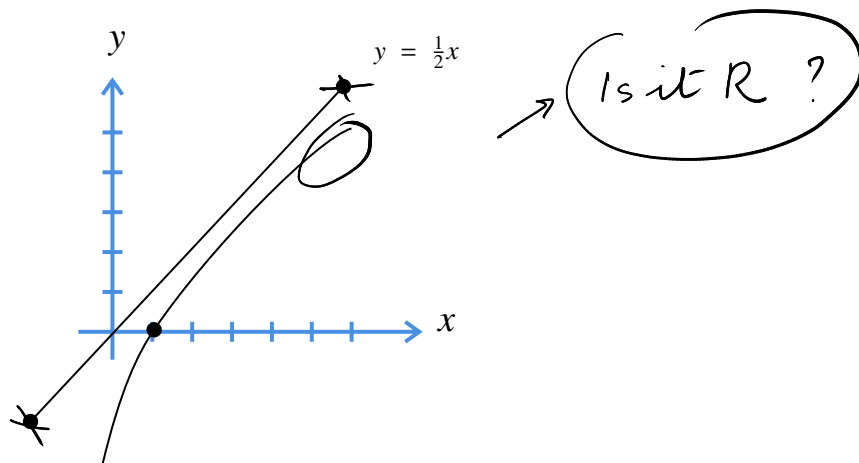
Let $A(u) = 1$ and $\log_A x = \int_1^x \frac{du}{1} = x - 1$ This is the ordinary log.

⁶For example 6.2 of Ay et al. [9] and references therein.

Example 3:-

$$\frac{d}{dx} \left\{ \frac{1}{2} \right\} \left\{ x - \frac{1}{x} \right\} = \frac{1}{2} \left\{ 1 + \frac{1}{x^2} \right\}$$

$$A(u) = \frac{2u^2}{1+u^2} \text{ and } \log_A x = \int_1^x \frac{du}{A(u)}$$



Example 4:-

$$\log_A x = \frac{x^{1-\alpha} - 1}{1-\alpha}, \alpha \neq 1, \text{ for } \alpha = \frac{1}{2}$$

$$\log_A x = 2\{x^{\frac{1}{2}} - 1\} = \int_1^x \frac{du}{\sqrt{u}}$$

Example 5:-

$$\log_A x = \frac{1}{2}(x + \log(x) - 1) = \int_1^x \frac{du}{\frac{2u}{1+u}}$$

1.10.1 Deformed Exponential

Let $\exp_A = \log_A^{-1}$ is convex function.

Example 1:-

$$\log(x) = y \text{ and } y = \exp(x)$$

Example 2:-

$$\log_A(x) = x - 1 \text{ and } \exp_A(y) = 1 + y$$

Example 3:-

$$\frac{1}{2} \left\{ x - \frac{1}{x} \right\} = y \iff x^2 - 2y^2 - 1 = 0 \text{ and } x > 0 \implies \exp_A(y) = y + \sqrt{1 + y^2}$$

Example 4:-

$$2(x^{\frac{1}{2}} - 1) = y, \quad \exp_A(y) = \left\{ 1 + \frac{1}{2}y^2 \right\}$$

Example 5:-

$$\frac{1}{2}(x + \log(x) - 1) = y \iff 1 + 2y = \log(xe^x)$$

From the

$$\exp_A(\log_A(x)) = x \implies \exp'_A(y) = A(\exp_A(y))$$

1.10.2 Exponential Family(Non Parametric)

Let us consider that X is finite space, P is probability function, and u is random variable then

$$q = \exp_A(u - K_p(u) + \log_A(p))$$

is a probability function and $K_p(u)$ is the solution of

$$\sum_x \exp_A(u(x) - \cancel{K_p(u)} + \log_A p(x)) = 1$$

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Now by computing the derivative $dK_p(u)[h] = \frac{d}{dt}K_p(u + th)|_{t=0}$

$$\begin{aligned}
 0 &= \sum_x \exp_A(u(x) - K_p(u) + \log_A p(x)) \{h(x) - dK_p(u)[h]\} \\
 &= \sum_x q(x) \{h(x) - dK_p(u)[h]\} \text{ that is} \\
 &= \boxed{\mathbb{E}_{\tilde{q}(h)=dK_p(u)[h]}} \text{ where } \tilde{q} = \left(\frac{A(q)}{\sum_x A(q(x))} \text{ add choose } \mathbb{E}_{\tilde{q}(u)=0} \right) ?
 \end{aligned}$$

1.10.3 Affine Spaces

Define the bundle of the probability function

$$\{(q, v) | q \in \mathcal{T}(X), \mathbb{E}(\tilde{v}) = 0\} \text{ and the charts}$$

$$s_p(q) = \log_A(q) - \log_A(p) - \mathbb{E}_{\tilde{p}}(\log_A(q) - \log_A(p))$$

Example 1:-

$$\begin{aligned}
 s_p(q) &= \frac{1}{2} \left\{ q - \frac{1}{q} \right\} - \frac{1}{2} \left\{ p - \frac{1}{p} \right\} - \\
 &\sum_x \frac{2p(x)^2}{1 + 2p(x)^2} \left\{ \frac{1}{2} \left\{ q(x) - \frac{1}{q(x)} \right\} - \frac{1}{2} \left\{ p(x) - \frac{1}{p(x)} \right\} \right\} \times \frac{1}{\sum_x \frac{2p(x)^2}{1 + 2p(x)^2}}
 \end{aligned}$$

1.10.4 Divergence

$$s_p^{-1}(u) = \exp_A(u - K_p(u) = \log_A(p)) \text{ and } u \in B_p$$

The deformed KL divergence will be

$$D(p \parallel q) = \mathbb{E}_{\tilde{q}}(\log_A(p) - \log_a(q))$$

Example 1:-

$$\sum_x \frac{2p(x)}{1 + p(x)} (p(x) - q(x)) + \log \frac{p(x)}{q(x)}$$

1.10.5 Score

The velocity is

$$\begin{aligned}\frac{d}{dt}s_p(q(H)) &= \frac{d}{dt}\left\{\log_A q(t) - \log_A(p) - \mathbb{E}_{\tilde{p}}(\log_A q(t) - \log_A(p))\right\} \\ &= \frac{\dot{q}(t)}{A(q(t))} - \mathbb{E}_{\tilde{q}}\frac{\dot{q}(t)}{A(q(t))}\end{aligned}$$

The score is the velocity of the moving frame $p = q(t)$

$$\begin{aligned}\frac{D}{dt}\left\{\frac{\dot{q}(t)}{A(q(t))} - \mathbb{E}_{\tilde{q}}\frac{\dot{q}(t)}{A(q(t))}\right\} &= \frac{\dot{q}(t)}{A(q(t))} \\ \mathbb{E}_{\tilde{q}}\frac{\dot{q}(t)}{A(q(t))} &= \sum \frac{q(x;t)}{A(q(x;t))} \frac{A(q(x;t))}{\sum_y A(q(y;t))} = 0\end{aligned}$$

Chapter 2

Kinematics and Linear Programming

Bibliography

- [1] Hermann Weyl. *Space- time- matter / by Hermann Weyl.* eng. translation of the 1921 RAUM ZEIT MATERIE. New York: Dover, 1952.
- [2] Nicolas Bourbaki. *Variétés différentielles et analytiques. Fascicule de résultats / Paragraphes 1 à 7.* Éléments de mathématiques XXXIII. Paris: Hermann, 1971.
- [3] Lev D. Landau and Eugenij M. Lifshits. *Course of Theoretical Physics. Statistical Physics.* 3rd. Vol. V. Butterworth-Heinemann, 1980.
- [4] I. A. Ibragimov and R. Z. Has minskii. *Statistical estimation.* Vol. 16. Applications of Mathematics. Asymptotic theory, Translated from the Russian by Samuel Kotz. Springer-Verlag, New York-Berlin, 1981, vii+403. ISBN: 0-387-90523-5.
- [5] Serge Lang. *Differential and Riemannian manifolds.* Third. Vol. 160. Graduate Texts in Mathematics. New York: Springer-Verlag, 1995, xiv+364. ISBN: 0-387-94338-2.
- [6] Alexander Barvinok. *A course in convexity.* Vol. 54. Graduate Studies in Mathematics. American Mathematical Society, Providence, RI, 2002, x+366. ISBN: 0-8218-2968-8.

Bibliography

- [7] Vera Pawlowsky-Glahn, Juan José Egozcue, and Raimon Tolosana-Delgado. *Modelling and Analysis of Compositional Data*. John Wiley & Sons, Ltd, Mar. 2015. DOI: 10.1002/9781119003144. URL: <https://doi.org/10.1002/9781119003144>.
- [8] Bradley Efron and Trevor Hastie. *Computer age statistical inference*. Vol. 5. Institute of Mathematical Statistics (IMS) Monographs. Algorithms, evidence, and data science. Cambridge University Press, New York, 2016, xix+475. ISBN: 978-1-107-14989-2. URL: <https://doi.org/10.1017/CB09781316576533>.
- [9] Nihat Ay et al. *Information geometry*. Vol. 64. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics [Results in Mathematics and Related Areas. 3rd Series. A Series of Modern Surveys in Mathematics]. Springer, Cham, 2017, xi+407. ISBN: 978-3-319-56477-7; 978-3-319-56478-4.
- [10] Jasminder S Sidhu and Pieter Kok. Geometric perspective on quantum parameter estimation. In: *AVS Quantum Science 2.1* (2020), 014701.
- [11] Jonas Erb and Nihat Ay. The Information-Geometric Perspective of Compositional Data Analysis. In: *Advances in Compositional Data Analysis: Festschrift in Honour of Vera Pawlowsky-Glahn*. Ed. by Peter Filzmoser et al. arXiv:2005.11510. Cham: Springer International Publishing, 2021, 21–43. ISBN: 978-3-030-71175-7. DOI: 10.1007/978-3-030-71175-7_2. URL: https://doi.org/10.1007/978-3-030-71175-7_2.
- [12] Goffredo Chirco and Giovanni Pistone. *Dually affine Information Geometry modeled on a Banach space*. arXiv:2204.00917. 2022.

Bibliography

DOI: 10.48550/ARXIV.2204.00917. URL: <https://arxiv.org/abs/2204.00917>.

- [13] Jiechao Zhang, Hadi Beik Mohammadi, and Leonel Rozo. Learning Riemannian Stable Dynamical Systems via Diffeomorphisms. In: *6th Annual Conference on Robot Learning*. 2022.